how fast an asset owner can invest before all others do the same. A better justification for investment, at least for slow-moving asset owners, is the persistence of a behavioral bias because there are barriers to the entry of capital. Some of these barriers may be structural, like the inability of certain investors to take advantage of this investment opportunity. Regulatory requirements, for example, force some investors to hold certain types of assets, like bonds above a certain credit rating or stocks with market capitalizations above a certain threshold. If there is a structural barrier to entry, then the behavioral bias can be exploited for a long time.

For some risk premiums, the most compelling explanations are rational (as with the volatility risk premium), for some behavioral (e.g., momentum), and for some others a combination of rational and behavioral stories prevails (like value/growth investing). Overall, the investor should not care if the source is rational or behavioral; the more appropriate question is whether she is different from the average investor who is subject to the rational or behavioral constraints and whether the source of returns is expected to persist in the future (at least in the short term). We take up this topic in detail in chapter 14, where I discuss factor investing.

7. The 2008–2009 Financial Crisis Redux

The simultaneously dismal performance of many risky assets during the financial crisis is consistent with an underlying multifactor model in which many asset classes were exposed to the same factors. The financial crisis was the quintessential bad time: volatility was very high, economic growth toward the end of the crisis was low, and there was large uncertainty about government and monetary policy. Liquidity dried up in several markets. The commonality of returns in the face of these factor risks is strong evidence in favor of multifactor models of risk, rather than a rejection of financial risk theory as some critics have claimed. Assets earn risk premiums to compensate for exposure to these underlying risk factors. During bad times, asset returns are low when these factor risks manifest. Over the long run, asset risk premiums are high to compensate for the low returns during bad times.

Some commentators have argued that the events of 2008 demonstrate the failure of diversification. Diversification itself is not dead, but the financial crisis demonstrated that asset class labels can be highly misleading, lulling investors into the belief that they are safely diversified when in fact they aren't. What matters are the embedded factor risks; assets are bundles of factor risks. We need to understand the factor risks behind assets, just as we look past the names and flavors of the things that we eat to the underlying nutrients to ensure we have enough to sustain us. We take on risk to earn risk premiums in the long run, so we need to understand when and how that factor risk can be realized in the short